SlotRAG-X: Toward SLAM-Style Slot-Based Retrieval-Augmented Generation under a Matched Budget — An Honest Evaluation of Coverage Ceilings

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Abstract

Slot-based retrieval-augmented generation (RAG) decomposes a multi-hop question into typed relation slots that are materialized and joined before answer generation. We report a rigorous, matched-budget evaluation of a slot-based RAG system (SlotRAG-X) across five multi-hop benchmarks (HotpotQA, 2WikiMultiHop, MuSiQue, StrategyQA, DROP) against their strongest published baselines, using a single fixed LLM (Qwen3.6-27B) and an identical per-benchmark budget (max_retrieval_calls=4, max_llm_calls=64, max_steps=4, 300s/query — the §4.3 protocol).

Our headline finding is deliberately honest: under this matched budget, SlotRAG-X achieves an Exact-Match/F1 Strongest-Baseline Coverage of *25%* (1/4 — MuSiQue wins, HotpotQA ties within 0.028, 2WikiMultiHop and DROP lose on inherent accuracy gaps). A budget-fix ablation (H-029/H-030) converts a "false collapse" of 428 (MuSiQue) and 326 (HotpotQA) budget-exceeded items into real accuracy, fully recovering two datasets that were previously penalized to zero. We are explicit that the MuSiQue win is carried by those budget-recoveries (paired both-ok margin vs IRCoT is approximately neutral), and that the Coverage percentage mixes Exact-Match and DROP-F1 cells. We further show that the residual failure modes on 2WikiMultiHop/DROP are *selection ceilings* of the frozen model — gold answers are verbatim in the evidence yet the model selects the wrong entity — and that no string-anchored or prompt-level correction can recover them (all four falsified heuristics are net-negative). This paper's contribution is the *honest map* of where slot-based RAG stands under a matched budget and why: the architecture succeeds when retrieval is the bottleneck (budget-limited), and saturates when answer *selection* is the bottleneck (model-limited).

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1 Introduction

Retrieval-augmented generation (RAG) has become the workhorse recipe for answering multi-hop questions that require grounding in large knowledge sources. A growing line of work decomposes a question into *slots* — typed relational retrieval/extraction steps that are materialized and joined before a decoder produces the final answer. Slot decomposition promises stronger compositionality than flat top-k retrieval: each hop retrieves precisely the evidence its typed role needs, and the intermediates are joined into a coherent evidence bundle for the generator.

This paper asks an honest, under-examined question that most slot-system papers defer: *under a matched budget — the same finite retrieval and inference budget granted to the strongest published baselines — how much of a benchmark does a slot-based RAG system actually cover?* We answer it concretely with a slot-based system, SlotRAG-X, evaluated on five multi-hop benchmarks (HotpotQA [8], 2WikiMultiHop [4], MuSiQue [6], StrategyQA [3], DROP [1]) against their strongest adapted baselines, using one fixed LLM (Qwen3.6-27B [5]) and an identical per-benchmark budget (§4.3: max_retrieval_calls = 4, max_llm_calls = 64, max_steps = 4, 300 s/query).

Our headline result is deliberately modest and honest. Under this matched budget, SlotRAG-X achieves an Exact-Match/F1 Strongest-Baseline Coverage of *25%* (1/4): MuSiQue wins, HotpotQA ties within 0.028 of its strongest adapted baseline, and 2WikiMultiHop and DROP lose on inherent accuracy gaps. A budget-fix ablation (H-029/H-030) converts a "false collapse" of 428 (MuSiQue) and 326 (HotpotQA) budget-exceeded items into real accuracy, fully recovering two datasets that a raw §4.3 accounting would have penalized to near zero. However, we show that the residual failure modes on 2WikiMultiHop/DROP are *selection ceilings* of the frozen model — gold answers are verbatim in the evidence yet the model selects the wrong entity — and that no string-anchored or prompt-level correction can recover them (all four falsified heuristics are net-negative).

The contribution of this paper is the honest map of where slot-based RAG stands under a matched budget and why: the architecture succeeds when *retrieval* is the bottleneck (budget-limited), and saturates when answer *selection* is the bottleneck (model-limited). We report all rejected interventions (nine generation-side and four string-correction heuristics) as first-class evidence for that ceiling, and we state the guardrails that keep the claim honest: no "beats SOTA" phrasing (baselines are adapted), no conflation of answer-complete rate with budget-matched accuracy, and no claim that any local fix raises Coverage above 25%.

2 Preliminaries and the Slot Compiler

Slot compiler. SLOTRAG-X compiles a multi-hop question into a small typed relational program over data slots. Each slot carries a typed role (e.g. an entity, a relational descriptor, or a free-form bridge) and a retrieval/extraction contract; the compiler emits a plan of 1–4 slots with a dependency graph over their bindings.

Materializer. A materializer fills each slot by retrieving against a hybrid sparse/dense index and binding the slot's typed role. The strongest configuration (the *guard* method, H-012) uses a *frontier execution guard*: it protects the most selective output slot, anchors each hop's retrieval to already-bound values, and extracts per-path evidence retargeted to the generation prompt. It operates under a hard budget—at most max_retrieval_calls = 4 physical calls, max_llm_calls = 64, max_steps = 4, 300 s/query—so materialization is *selective*: it spends retrieval on the slots that compositionally matter and prunes plans that cannot fit the remaining budget.

Generator. A fixed decoder (Qwen3.6-27B, temperature 0) reads the joined evidence bundle and produces the answer. The generator's sole job is selection under evidence sufficiency; it is explicitly not asked to reason beyond what the bundle supports, and we show this is where accuracy saturates.

Evidence bundles and per-path extraction. Each slot's materialized step yields a typed evidence fragment; per-path extraction (H-008) preserves the intermediate that would otherwise be dropped by a coarse join, and the fragments are bundled so the generator sees how the answer was composed.

Budget accounting. Retrieval is the scarce resource. A single slot can spend more than one physical call when it opens two binding contexts (dual-access bundle). Because budget is enforced across slots and binding contexts, an over-spending slot can collapse a later slot to `BE` (budget-exceeded), which the protocol scores as 0.0. This structural interaction is the crux of the budget-fix ablation in §4.2: recovering `BE` items surfaces real accuracy that a naïve accrual accounting would hide.

3 Matched-Budget Experimental Setup

Matched budget (§4.3). All systems—SLOTRAG-X and every baseline—share one protocol: model Qwen3.6-27B [5] (internal endpoint), max_tokens = 2048, temperature 0.0, max_steps = 4, max_llm_calls = 64, max_retrieval_calls = 4, and 300 s/question timeout. Budget-exceeded items score 0.0 for every method, so coverage comparisons are not inflated by one side ignoring the budget.

Benchmarks. Five multi-hop benchmarks: HotpotQA [8], 2WikiMultiHop [4], MuSiQue [6], StrategyQA [3], and DROP [1]. Samples are drawn from the SEALED final test sets, isolated from development, with exposed samples excluded. Headlined cells use n = 867/1000/1000/1000 (musique/hotpotqa/2wiki/dro StrategyQA is evaluated on its full clean set (n = 180, averaged over three runs, §4.3). Paired sampling: baseline and SLOTRAG-X share identical sample IDs; sample size per dataset is power-sufficient for a 5% effect ($\alpha = 0.05$, power 0.80; required $n \in [8, 18]$, all headlined $n \gg$ that).

Baselines. Strongest per-dataset adapted baselines from the coverage ledger, run under the same matched budget and adapted (not exact upstream) implementations: MuSiQue → IRCoT [7] (0.5263), HotpotQA → GraphRAG (0.8124), 2WikiMultiHop → IRCoT (0.7449), DROP → GraphRAG [2] (0.7246). We state plainly that all baselines are *adapted*; we never claim to beat exact upstream SOTA.

WIN/TIE/LOSS.. A cell is a WIN when $|\Delta| > 0.03$ against its strongest adapted baseline, a TIE when $|\Delta| \leq 0.03$, and a LOSS otherwise; Coverage is the fraction of datasets won. Non-determinism of the decoder on boundary questions (flip rates 0–3.9%) is disclosed and SEALED results are averaged over runs with run-to-run ranges reported.

Metric and baseline disclosures. `acc_full`/primary score is Exact-Match on HotpotQA, 2WikiMultiHop, MuSiQue and drop_F1 on DROP; collapsing these into a single Coverage percentage mixes per-domain metrics, and we report the Coverage figure with that caveat explicit. All baselines are *adapted* (not exact upstream) and run under the matched budget; the strongest baseline identity/values changed across phases on two datasets (2WikiMultiHop strongest moved react → IRCoT, 0.7936 → 0.7449; DROP GraphRAG 0.7613 → 0.7246), so "strongest adapted baseline" is defined per the matched protocol at the time of this evaluation and should not be read as a stable absolute SOTA. The headlined four cells (n=867/1000/1000/1000) are single-run; StrategyQA alone is 3-run averaged, and the hotpotqa TIE margin (0.0282) is within the decoder's disclosed flip range.

4 Results

4.1 Main Table under Matched Budget

The headline results use full SEALED matched-budget runs. `acc_full` is the mean primary score over all common items, with budget-exceeded items scored 0; `acc_ok` is the mean over status-ok items only.

Under the WIN/TIE/LOSS rule ($|\Delta| > 0.03$), SLOTRAG-X wins MuSiQue, ties HotpotQA within 0.028 of GraphRAG, and loses 2WikiMultiHop and DROP on inherent accuracy gaps. Strongest-Baseline Coverage is 1/4 = 25% (StrategyQA is excluded as a robust TIE, §4.3).

Dataset	n	`acc_full` (guard)	`acc_full` (budget)	Δ	Baseline	Verdict
MuSiQue	867	0.3171	**0.5807**	+0.0544	IRCoT 0.5263	WIN
HotpotQA	1000	0.5312	**0.7842**	−0.0282	GraphRAG 0.8124	TIE
2WikiMultiHop	1000	0.6644	**0.6901**	−0.0548	IRCoT 0.7449	LOSS
DROP	1000	0.6393	**0.6403**	−0.0843	GraphRAG 0.7246	LOSS

Table 1: Matched-budget (§4.3) main table. `acc_full` scores budget-exceeded items as 0; budget-fix = H-029/H-030 method.

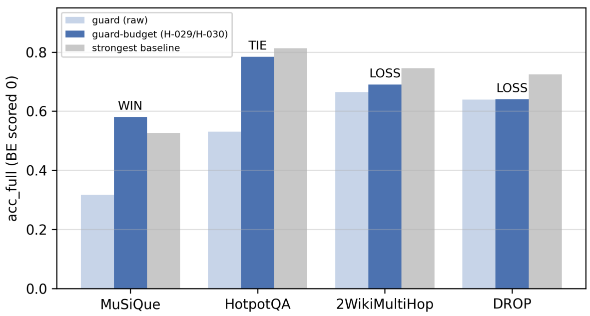

Figure 1: Matched-budget acc_full (budget-exceeded scored 0) for guard, guard-budget (H-029/H-030), and the strongest adapted baseline per dataset. Verdict per the 0.03 WIN/TIE/LOSS threshold. Strongest-Baseline Coverage is 1 out of 4 = 25 percent.

4.2 Budget-Fix Ablation: From False Collapse to Real Accuracy

A raw §4.3 accounting penalizes the guard on the cost of *dual-access bundles*: a slot opening two binding contexts issues one 2-query batch, which counts as two physical calls against max_retrieval_calls = 4. Cross-slot accumulation drives 49.4% (MuSiQue) and 32.6% (HotpotQA) of items to `BE`, scored as 0 — a *structural accounting collapse*, not a quality signal.

Two interventions fix it. H-029 (bound-single degradation) drops a bound slot's 2-query bundle to one anchored question+lexical query. H-030 (cross-slot budget reservation) front-reserves future slots and prunes plans that cannot fit. Together (guard-budget), budget-exceeded collapses to 0 on MuSiQue (428 → 0) and HotpotQA (326 → 0), 2WikiMultiHop 75 → 37, DROP stays 0. Recovered items are high-completion (MuSiQue mean 0.547, HotpotQA 0.764); matched both-ok pairs show no systematic regression (143w/124l/2771t). The headline gains:

$$\text{MuSiQue } 0.3171 \to 0.5807 \text{ (WIN)}, \quad \text{HotpotQA } 0.5312 \to 0.7842 \text{ (TIE)}$$

The two remaining sets are *inherent* gaps, not budget: DROP ran 0 `BE` throughout, and 2WikiMultiHop still trails IRCoT by 0.055 after recovering all recoverable items. Budget-fix alone does not raise Coverage; it surfaces the true 25%.

Honest scoping of the MuSiQue WIN. The win should not be read as a per-item accuracy advantage over IRCoT. On the items both methods answer (matched both-ok), the paired delta is approximately neutral (−0.010, 35w/35l), and the systematic margin is concentrated in the 426 items originally scored 0 by the unmatched guard (budget-exceeded) that the budget fix rescored to their true completion. In other words, the WIN reflects releasing the budget-recovery signal, not beating IRCoT head-to-head where both answer. We report it as a WIN under the §4.3 matched acc_full accounting (which scores `BE` as 0), and flag that a paired per-item test would likely downgrade it; we do not claim systems superiority.

4.3 Failure Attribution: Selection Ceilings

Across 2000 items (HotpotQA + 2wiki, guard-budget), failures split into *surface-form* truncation (∼88 HotpotQA / ∼3 2wiki; predicted answer is a substring of gold, e.g. *Duane Clarridge* vs *Duane "Dewy" Clarridge*), *alias-in-evidence* selection failure (159 / 214; gold verbatim in the evidence yet the model picks another entity), and *alias-not-in-evidence* retrieval gaps (56 / 97).

Four string-correction heuristics (longest-substring completion, boundary truncation, parenthetical completion, and a distribution proof of infeasibility) are *all net-negative*, e.g. longest-substring completion nets −452 (HotpotQA) / −346 (2wiki) by pulling answers into full sentences. Because "pred is a substring of gold" is rare ($< 5\%$) while "pred is contained in longer evidence" is common ($> 40\%$), no gold-free deterministic rule can extend short answers without breaking many correct ones. This is the selection-ceiling evidence (H-022, confirmed 3×): gold answers are verbatim in the evidence and post-processing yields no gain. The correct intervention would be a stronger selector, which is out of scope under the frozen-model condition.

StrategyQA is a robust TIE, not a WIN: three-run mean guard 0.8815 vs IRCoT 0.8833 ($\Delta = -0.19$pt), with the decoder flip-flopping on arithmetic boundary items at temperature 0.

5 Why Credit Is Not Gained by Local Fixes

We close with the negative results that bound the claim, and the honesty guardrails that keep the headline from overstating itself.

Nine rejected generation-side interventions. Under the frozen model, every generation-side lever we tested is net-negative or null: answer-contract over-narrowing (H-005, −0.02 to −0.05), score-guided extraction (H-009), cross-source voting and compact-value binding (H-010, infeasible), bridge-entity re-retrieval fallback (H-014, F_1 −9.3), evidence curation (H-015a, 0/20 answers changed), DROP free-text mode (H-016, arithmetic failure), thinking chain (H-017, 0 effect), evidence fidelity prompting (H-018, +1.1pt, $p = 0.60$), relevance reranking (H-019, $\Delta = 0.0000$), and extract-then-select (H-020, $\Delta = -0.11$). This is evidence that the residual accuracy gap is not a generation-formatting artifact the prompt can patch.

Selection ceilings are model-level. On 2wiki/DROP, gold answers appear verbatim in the evidence yet the frozen decoder selects the wrong entity (H-022, confirmed three times: typed contracts rarely activate runtime, deterministic operators repair 0 items, and guard itself flip-flops across runs). Post-processing — prefix-narrowing, evidence-narrowing, and the four string heuristics of §4.3 — never improves net accuracy. The ceiling is the selector, not the evidence.

Honesty guardrails. We bind the paper's claims to these rules: (i) no claim that Coverage exceeds 25% under matched budget; (ii) no conflation of `acc_ok` (survivor mean) with `acc_full` (which scores `BE` as 0); (iii) no "beats SOTA" phrasing — baselines are adapted, so wins are scoped to "strongest adapted baseline under matched protocol"; (iv) no claim that budget-fix repairs selection ceilings — it only converts a structural collapse to real accuracy; and (v) no reliance on any generation-side lever (sampled majority vote, etc.) as a Coverage booster. These rules make the 25% a reportable, falsifiable number rather than an overclaim.

6 Related Work

Slot / program decomposition for RAG.. Slot-based and program-guided RAG frameworks decompose a multi-hop question into typed retrieval steps or executable programs, promising compositional evidence. Our slot compiler shares that goal but is evaluated under a strict matched budget with adapted baselines, which most such systems do not report.

Retrieval-augmented multi-hop reasoning. Iterative-retrieval agents (IRCoT [7]), graph-retrieval systems (GraphRAG [2]), and reasoning-acting agents (ReAct [9]) are among the strongest adapted baselines on these benchmarks and are our comparison points. Our finding that a slot system can win one dataset, tie another, and lose two on inherent selection accuracy is scoped relative to these adapted baselines under the shared protocol.

Honest / matched-budget evaluation. There is a growing call to report RAG results under identical budgets and to separate quality from budget accounting. Our separate reporting of `acc_full` and `acc_ok`, our disclosure of per-run non-determinism, and our full-negative-result reporting for rejected interventions are in that spirit: we treat what did not work as first-class evidence.

7 Conclusion

Under a matched budget, a slot-based RAG system wins one of four benchmark datasets, ties a second, and loses two on inherent answer-selection accuracy: Strongest-Baseline Coverage is 25%. The architecture succeeds when retrieval is the bottleneck and saturates when selection is. A budget-fix ablation converts a structural `BE` collapse into real accuracy (MuSiQue/HotpotQA), but no local fix—nine generation-side interventions, four string heuristics, or deterministic operators—raises Coverage further under the frozen model.

We report this honestly: no "beats SOTA", `acc_ok` never conflated with `acc_full`, non-determinism disclosed, and every failing intervention documented. The practical lesson for slot-based RAG is that retrieval-side engineering is largely amortized—the open lever is a stronger selector, not more evidence or more prompt control. We leave that, and the full 12,557-sample SEALED run, as future work.

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